MASTER'S THESIS

Towards a WasmGC backend and runtime for the MicroHs compiler

Author:
Martijn Voorwinden

Supervisor:
Dr. Marco Vassena

Second examiner:
Dr. Wouter Swierstra

Daily supervisor:
Apoorva Anand MSc.

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Utrecht University

Sharing science,
shaping tomorrow

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The acknowledgments and the people to thank go here, don't forget to include your project advisor…

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Chapter 1

Chapter Title Here

1.1 Welcome and Thank You

Welcome to this LaTeX Thesis Template, a template for writing a thesis using the LaTeX typesetting system.

If you are writing a thesis (or will be in the future) and its subject is technical, then creating it in LaTeX is highly recommended as a way to make sure you can just get down to the essential writing without having to worry over formatting or wasting time arguing with your word processor.

LaTeX is easily able to professionally typeset documents that run to hundreds or thousands of pages long. With simple mark-up commands, it automatically sets out the table of contents, margins, page headers and footers and keeps the formatting consistent and beautiful. One of its main strengths is the way it can easily typeset mathematics, even *heavy* mathematics. Even if those equations are the most horribly twisted and most difficult mathematical problems that can only be solved on a super-computer, you can at least count on LaTeX to make them look stunning.

1.2 Learning LaTeX

LaTeX is not a WYSIWYG (What You See is What You Get) program, unlike word processors such as Microsoft Word or Apple's Pages. Instead, a document written for LaTeX is actually a simple, plain text file that contains *no formatting*. You tell LaTeX how you want the formatting in the finished document by writing in simple commands amongst the text, for example, if I want to use *italic text for emphasis*, I write the `\emph{text}` command and put the text I want in italics in between the curly braces. This means that LaTeX is a "mark-up" language, very much like HTML.

1.2.1 A (not so short) Introduction to LaTeX

If you are new to LaTeX, there is a very good eBook – freely available online as a PDF file – called, "The Not So Short Introduction to LaTeX". The book's title is typically shortened to just *lshort*. You can download the latest version (as it is occasionally updated) from here: http://www.ctan.org/tex-archive/info/lshort/english/lshort.pdf

1.2.2 A Short Math Guide for LaTeX

If you are writing a technical or mathematical thesis, then you may want to read the document by the AMS (American Mathematical Society) called, "A Short Math Guide for LaTeX". It can be found online here: http://www.ams.org/tex/amslatex.html under the "Additional Documentation" section towards the bottom of the page.

1.2.3 Common LaTeX Math Symbols

There are a multitude of mathematical symbols available for LaTeX and it would take a great effort to learn the commands for them all. The most common ones you are likely to use are shown on this page: http://www.sunilpatel.co.uk/latex-type/latex-math-symbols/

You can use this page as a reference or crib sheet, the symbols are rendered as large, high quality images so you can quickly find the LaTeX command for the symbol you need.

1.2.4 LaTeX on a Mac

The LaTeX distribution is available for many systems including Windows, Linux and Mac OS X. The package for OS X is called MacTeX and it contains all the applications you need – bundled together and pre-customized – for a fully working LaTeX environment and work flow.

MacTeX includes a custom dedicated LaTeX editor called TeXShop for writing your '`.tex`' files and BibDesk: a program to manage your references and create your bibliography section just as easily as managing songs and creating playlists in iTunes.

1.3 Getting Started with this Template

If you are familiar with LaTeX, then you should explore the directory structure of the template and then proceed to place your own information into the *THESIS INFORMATION* block of the `main.tex` file. You can then modify the rest of the file to add your name, the names of your supervisors, and title of your thesis.

If you are new to LaTeX it is recommended that you carry on reading through the rest of the information in this document.

1.3.1 About this Template

This LaTeX Thesis Template is originally based and created around a LaTeX style file created by Steve R. Gunn from the University of Southampton (UK), department of Electronics and Computer Science. You can find his original thesis style file at his site, here: http://www.ecs.soton.ac.uk/~srg/softwaretools/document/templates/

Steve's `ecsthesis.cls` was then taken by Sunil Patel who modified it by creating a skeleton framework and folder structure to place the thesis files in. The resulting template can be found on Sunil's site here: http://www.sunilpatel.co.uk/thesis-template

Sunil's template was made available through http://www.LaTeXTemplates.com where it was modified many times based on user requests and questions. Version 2.0 and onwards of this template represents a major modification to Sunil's template and is, in fact, hardly recognisable. The work to make version 2.0 possible was carried out by Vel and Johannes Böttcher.

The template and thesis class was later adapted by Wouter Swierstra, adding the specific information relevant for students at Utrecht University.

1.4 What this Template Includes

1.4.1 Folders

This template comes as a single zip file that expands out to several files and folders. The folder names are mostly self-explanatory:

chapters – this is the folder where you put the thesis chapters. You may want to split each chapter in its own separate `.tex` file and they can be split as:

- Chapter 1: Introduction to the thesis topic
- ...
- Chapter N: Conclusion and future directions

img – this folder contains all images for the thesis. It's a bit cleaner to store images in a separate directory.

1.4.2 Files

Included are also several files, most of them are plain text and you can see their contents in a text editor. After initial compilation, you will see that more auxiliary files are created by LaTeX or BibTeX and which you don't need to delete or worry about:

main.bib – this is an important file that contains all the bibliographic information and references that you will be citing in the thesis for use with BibTeX. You can write it manually, but there are reference manager programs available that will create and manage it for you. Bibliographies in LaTeX are a large subject and you may need to read about BibTeX before starting with this. Many modern reference managers will allow you to export your references in BibTeX format which greatly eases the amount of work you have to do.

thesis.cls – this is an important file. It is the class file that tells LaTeX how to format the thesis.

main.pdf – this is your beautifully typeset thesis (in the PDF file format) created by LaTeX. It is supplied in the PDF with the template and after you compile the template you should get an identical version.

main.tex – this is an important file. This is the file that you tell LaTeX to compile to produce your thesis as a PDF file. It contains the framework and constructs that tell LaTeX how to layout the thesis. It is heavily commented so you can read exactly what each line of code does and why it is there. After you put your own information into the *THESIS INFORMATION* block – you have now started your thesis!

Files that are *not* included, but are created by LaTeX as auxiliary files include:

main.aux – this is an auxiliary file generated by LaTeX, if it is deleted LaTeX simply regenerates it when you run the main `.tex` file.

main.bbl – this is an auxiliary file generated by BibTeX, if it is deleted, BibTeX simply regenerates it when you run the `main.aux` file. Whereas the `.bib` file contains all the references you have, this `.bbl` file contains the references you have actually cited in the thesis and is used to build the bibliography section of the thesis.

main.blg – this is an auxiliary file generated by BibTeX, if it is deleted BibTeX simply regenerates it when you run the main `.aux` file.

main.lof – this is an auxiliary file generated by LaTeX, if it is deleted LaTeX simply regenerates it when you run the main `.tex` file. It tells LaTeX how to build the *List of Figures* section, if you are using this.

main.log – this is an auxiliary file generated by LaTeX, if it is deleted LaTeX simply regenerates it when you run the main `.tex` file. It contains messages from LaTeX, if you receive errors and warnings from LaTeX, they will be in this `.log` file.

main.lot – this is an auxiliary file generated by LaTeX, if it is deleted LaTeX simply regenerates it when you run the main `.tex` file. It tells LaTeX how to build the *List of Tables* section, if you are using this.

main.out – this is an auxiliary file generated by LaTeX, if it is deleted LaTeX simply regenerates it when you run the main `.tex` file.

So from this long list, only the files with the `.bib`, `.cls` and `.tex` extensions are the most important ones. The other auxiliary files can be ignored or deleted as LaTeX and BibTeX will regenerate them.

There is a very rudimentary `Makefile` included to build the thesis, including the bibliography, and delete any of these generated files.

1.5 Filling in Your Information in the `main.tex` File

You will need to personalise the thesis template and make it your own by filling in your own information. This is done by editing the `main.tex` file in a text editor or your favourite LaTeX environment.

Open the file and scroll down to the third large block titled *THESIS INFORMATION*. Fill out the information about yourself, your supervisor, and your thesis title.

1.6 The `main.tex` File Explained

The `main.tex` file contains the structure of the thesis. There are plenty of written comments that explain what pages, sections and formatting the LaTeX code is creating. Each major document element is divided into commented blocks with titles in all capitals to make it obvious what the following bit of code is doing. Initially there seems to be a lot of LaTeX code, but this is all formatting, and it has all been taken care of so you don't have to do it.

Begin by checking that your information on the title page is correct.

Next come the acknowledgements. On this page, write about all the people who you wish to thank (not forgetting parents, partners and your advisor/supervisor).

Finally, there is the block where the chapters are included. Uncomment the lines (delete the % character) as you write the chapters. Each chapter should be written in its own file and put into the *chapters* folder and named `chapter1`, `chapter2`, etc...Similarly for the appendices, uncomment the lines as you need them. Each appendix should go into its own file, also in the *chapters* folder.

After the preamble, chapters and appendices finally comes the bibliography. The bibliography style (called *`authoryear`*) is used for the bibliography and is a fully featured style that will even include links to where the referenced paper can be found online. If you want to use a different citation style, you may want to change the bibliography style. Check with your supervisor which style of bibliography is the most common in your area of research. Do not underestimate how grateful your reader will be to find that a reference to a paper is just a click away. Of course, this relies on you putting the URL information into the BibTeX file in the first place.

1.7 Thesis Features and Conventions

To get the best out of this template, there are a few conventions that you may want to follow.

One of the most important (and most difficult) things to keep track of in such a long document as a thesis is consistency. Using certain conventions and ways of doing things (such as using a Todo list) makes the job easier. Of course, all of these are optional and you can adopt your own method.

1.7.1 Printing Format

The headers for the pages contain the page number on the outer side (so it is easy to flick through to the page you want) and the chapter name on the inner side.

The text is set to 11 point by default with single line spacing, again, you can tune the text size and spacing should you want or need to using the options at the very start of `main.tex`.

1.7.2 References

The `biblatex` package is used to format the bibliography and inserts references to papers. To insert citations, use the `citep` command, e.g. [1]. The default options used in the `main.tex` ensure that citations of references are formatted with the author(s) listed with the date of the publication. If a paper has many different authors, BibTeX will abbreviate all the authors after the first as so: [2].

If you want to cite more than one paper at once, you can use the `citep` command to do so, separating references in the `tex` source with semicolons (e.g. [1, 2]).

To see how you use references, have a look at the `Chapter1.tex` source file. You can customize the bibliography style – there are different ways to include citations in your text. For example, you can also write: **(author?)** [1] did amazing work; or only the display the year of publication **(year?)**. There is a good cheatsheet available online.

Check with your supervisor how citations are typically given in your area of research; sometimes numbered citations are more common. If that is the case, be sure to change the bibliography style specified in `main.tex` file to:

```
\bibliographystyle{plain}
```

Scientific references should come *before* the punctuation mark if there is one (such as a comma or period). That is, write a citation like so [1]. Never write a citation after a punctuation mark, like this.[1]

Try to never use citations as a noun. It is fine to write "**(author?)** [1] presents an important result about treewidth" – but never write "an important result about treewidth is found in [1]" – if you use a numerical bibliography style this is typically as something like "an important result about treewidth is found in [1]" which is very hard to understand as a reader!

The bibliography is typeset with references listed in alphabetical order by the first author's last name. This is similar to the APA referencing style. To see how LaTeX typesets the bibliography, have a look at the very end of this document (or just click on the reference number links in in-text citations).

1.7.3 Tables

Tables are an important way of displaying your results, below is an example table which was generated with this code:

```
\begin{table}
\caption{The effects of treatments X and Y on the four groups studied.}
\label{tab:treatments}
\centering
\begin{tabular}{l l l}
\toprule
\tabhead{Groups} & \tabhead{Treatment X} & \tabhead{Treatment Y} \\
\midrule
1 & 0.2 & 0.8\\
2 & 0.17 & 0.7\\
3 & 0.24 & 0.75\\
4 & 0.68 & 0.3\\
\bottomrule\\
\end{tabular}
\end{table}
```

TABLE 1.1: The effects of treatments X and Y on the four groups studied.

Groups	Treatment X	Treatment Y
1	0.2	0.8
2	0.17	0.7
3	0.24	0.75
4	0.68	0.3

You can reference tables with `\ref{<label>}` where the label is defined within the table environment. See `Chapter1.tex` for an example of the label and citation (e.g. Table 1.1).

1.7.4 Figures

There will hopefully be many figures in your thesis (that should be placed in the *img* folder). The way to insert figures into your thesis is to use a code template like this:

```
\begin{figure}
  \begin{center}
  \includegraphics[scale=0.5]{img/logo.jpg}
  \end{center}
  \caption{The UU logo}
  \label{fig:uu-logo}
\end{figure}
```

FIGURE 1.1: The UU logo

Also look in the source file. Putting this code into the source file produces the picture of the UU logo, scaled down by 50%.

Sometimes figures don't always appear where you write them in the source. The placement depends on how much space there is on the page for the figure. Sometimes there is not enough room to fit a figure directly where it should go (in relation to the text) and so LaTeX puts it at the top of the next page. Positioning figures is the job of LaTeX and so you should only worry about making them look good!

Figures usually should have captions just in case you need to refer to them (such as in Figure 1.1). The `\caption` command contains two parts, the first part, inside the square brackets is the title that will appear in the *List of Figures*, and so should be short. The second part in the curly brackets should contain the longer and more descriptive caption text.

LaTeX is capable of using images in pdf, jpg and png format.

1.7.5 Typesetting mathematics

If your thesis is going to contain heavy mathematical content, be sure that LaTeX will make it look beautiful, even though it won't be able to solve the equations for you.

The "Not So Short Introduction to LaTeX" (available on CTAN) should tell you everything you need to know for most cases of typesetting mathematics. If you need more information, a much more thorough mathematical guide is available from the AMS called, "A Short Math Guide to LaTeX" and can be downloaded from: `ftp://ftp.ams.org/pub/tex/doc/amsmath/short-math-guide.pdf`

There are many different LaTeX symbols to remember, luckily you can find the most common symbols in The Comprehensive LaTeX Symbol List.

You can write an equation, which is automatically given an equation number by LaTeX like this:

```
\begin{equation}
P(A \mid B) = \frac{P(B \mid A)\, P(A)}{P(B)}
\label{eqn:Bayes}
\end{equation}
```

This will produce Bayes' theorem about probabilities:

$$P(A \mid B) = \frac{P(B \mid A)\, P(A)}{P(B)} \tag{1.1}$$

All equations you write (which are not in the middle of paragraph text) are automatically given equation numbers by LaTeX. Using the `label` and `ref` you can refer to the equation number; for example, Bayes' theorem above is numbered 1.1.

If you don't want a particular equation numbered, use the unnumbered form:

```
\[ a^{2}=4 \]
```

The AMS Math package has several other environments that are very useful for writing mathematics. For example, you might want to define the notion of *graph*:

Definition 1 (Graph)**.** *A* graph *is an ordered pair*

$$G = (V, E),$$

where V is a non-empty finite set whose elements are called vertices *(or* nodes*), and $E \subseteq \{\{u, v\} \mid u, v \in V,\ u \neq v\}$ is a set of unordered pairs of distinct elements of V, called* edges*.*

If the elements of E are ordered pairs, that is,

$$E \subseteq V \times V,$$

then G is called a directed graph *(or* digraph*).*

You read much more about producing such definitions and theorems in the documentation of the AMS math packages:

https://www.ams.org/arc/tex/amscls/amsthdoc.pdf

Once again, you can reference definitions that have a `label` command associated with them: graphs are defined in Definition 1.

1.7.6 Typesetting code

LaTeX has several options for typesetting code fragments.

The easiest (and ugliest option) is using the `verbatim` enviroment. This type sets any code fragment using a monowidth font, without any syntax highlighting.

```
# A Python script to print the string "Hello world"
print{"Hello world")
```

Nowadays, there are more advanced typesetting packages for typesetting syntax highligted code, such as `minted`:

https://tug.ctan.org/macros/latex/contrib/minted/minted.pdf

Alternatively, there are pre-processors such as `lhs2TeX` to produce typeset Haskell code:

https://www.andres-loeh.de/lhs2tex/

Instead of a LaTeX package, this is a pre-processor that produces LaTeX code. It is customizable and may be used to produce a variety different styles.

1.8 Sectioning and Subsectioning

You should break your thesis up into nice, bite-sized sections and subsections. LaTeX automatically builds a table of Contents by looking at all the `\chapter{}`, `\section{}` and `\subsection{}` commands you write in the source.

The Table of Contents should only list the sections to three (3) levels. A `chapter{}` is level zero (0). A `\section{}` is level one (1) and so a `\subsection{}` is level two (2). In your thesis it is likely that you will even use a `subsubsection{}`, which is level three (3). The depth to which the Table of Contents is formatted is set within `thesis.cls`. If you need this changed, you can do it in `main.tex`.

By adding a `label` command to the start of a section, you can refer to it easily later. For example, this is Section 1.8. Have a look at the `.tex` source to understand how this works.

Bibliography

[1] Hans L. Bodlaender. A linear-time algorithm for finding tree-decompositions of small treewidth. *SIAM Journal on Computing*, 25(6):1305–1317, 1996.

[2] Manuel M. T. Chakravarty, Roman Leshchinskiy, Simon Peyton Jones, Gabriele Keller, and Simon Marlow. Data parallel haskell: a status report. In *DAMP '07: Proceedings of the 2007 workshop on Declarative aspects of multicore programming*, pages 10–18. ACM, 2007.

Appendix A

Frequently Asked Questions

A.1 How do I change the colors of links?

The color of links can be changed to your liking using:

`\hypersetup{urlcolor=red}`, or
`\hypersetup{citecolor=green}`, or
`\hypersetup{allcolor=blue}`.

If you want to completely hide the links, you can use:

`\hypersetup{allcolors=.}`, or even better:
`\hypersetup{hidelinks}`.

If you want to have obvious links in the PDF but not the printed text, use:

`\hypersetup{colorlinks=false}`.